Texas College of Management & IT (TCMIT)

Faculty: Bachelor in Computer Science (BIT)

Semester: 3rd Semester

Course Title: Electronic Commerce

Assignment Type: Individual Assignment

Total Marks: 100

Weightage: As per course assessment scheme

Submission Mode: Hard Copy / LMS Upload / Live Project

PART A

Assignment Objectives:

1. Demonstrate critical understanding of e-commerce concepts
2. Apply theoretical frameworks to real-world and Nepalese contexts
3. Analyze technical, ethical, security, and strategic issues in e-commerce
4. Develop analytical, evaluative, and problem-solving skills

Assignment Instructions

1. Attempt ALL questions.
2. Each question carries 8 marks.
3. Answers must reflect critical thinking, not descriptive writing.
4. Use relevant theories, models, and examples wherever applicable.
5. Nepalese context must be emphasized where required.
6. Plagiarism beyond acceptable academic limits will be penalized.
7. Use proper citations and references.

Assignment Questions

Question 1: Fundamentals of Electronic Commerce (8 Marks)

Electronic commerce is often presented as a natural evolution of traditional commerce driven by digital technologies.Critically analyze whether e-commerce represents a fundamental transformation of business models or merely a digitized extension of traditional commerce. In your answer, evaluate:

- Differences in value creation, customer relationships, and cost structures between traditional and e-commerce models
- The role of network effects, platform economies, and data-driven decision-making
- Situations where e-commerce may fail to outperform traditional commerce, particularly in developing economies

Support your arguments with theoretical frameworks and real-world examples.

Question 2: Security Threats to E-Commerce (8 Marks)

Despite heavy investment in cybersecurity infrastructure, many e-commerce platforms remain vulnerable to breaches that cause financial and reputational damage.

Critically analyze whether e-commerce security failures are primarily technical, organizational, or strategic in nature. In your response:

- Evaluate the limitations of encryption, authentication, and automated fraud detection in real-world implementation
- Analyze how human behavior, organizational culture, and third-party dependencies undermine technical security controls
- Design a layered security governance model suitable for SMEs operating e-commerce platforms in developing countries

Question 3: Electronic Payment System (8 Marks)

Electronic payment systems are central to the success of e-commerce but vary widely in adoption and trust levels. Evaluate critically how electronic payment systems influence consumer trust, transaction efficiency, and market expansion. Your answer should:

- Compare traditional banking-based payment systems with digital wallets and mobile payments
- Analyze the impact of system failures, fraud, and transaction reversibility on user confidence
- Discuss whether payment innovation alone can drive e-commerce growth without parallel improvements in infrastructure and regulation
- Use contextual examples to support your reasoning.

Question 4: Ethical and Tax Issues of E-Commerce in Nepal (8 Marks)

The rapid growth of e-commerce in Nepal has raised significant ethical and regulatory challenges. Critically examine the ethical and taxation issues associated with e-commerce in Nepal by addressing:

- Consumer privacy, data protection, and transparency concerns
- Tax evasion, informal digital businesses, and cross-border transactions
- The challenges faced by regulatory bodies in enforcing existing laws
- Propose policy-level recommendations that balance innovation, consumer protection, and government revenue.

Question 5: Strategy for Marketing, Sales, and Promotion through E-Commerce (8 Marks)

E-commerce firms increasingly rely on data-driven digital marketing to gain competitive advantage, yet many struggle with declining engagement and rising acquisition costs.
Critically analyze whether digital marketing in e-commerce creates sustainable value or short-term performance illusions. In your answer:

- Evaluate the long-term effectiveness of algorithmic targeting, influencer economies, and behavioral analytics
- Analyze risks arising from over-reliance on third-party platforms, changing algorithms, and consumer data exploitation
- Design a customer-centric e-commerce marketing strategy focused on trust, loyalty, and lifetime value rather than immediate conversions
- Justify why ethical marketing may become a competitive necessity rather than a compliance requirement.

PART B (Mini Project)

Title: Building an E-commerce Website Using WordPress and WooCommerce
Project Overview
Type: Individual Project
Scope: Covers theoretical and practical applications of concepts from Chapters 1 to 10.

Objective: The objective of this assignment is to help students understand how to create a fully functional e-commerce website using WordPress and WooCommerce. Students will also learn how to enforce user authentication (login/signup) before checkout, set up the website on localhost, and then transfer it to a live server using cPanel.

Assignment Details: [60 Marks]

<u>Part 1: Setting Up the E-commerce Website on Localhost</u>

1. **Install WordPress Locally:**
    - Download and install a local server environment like WP-Local or XAMPP.
    - Set up a new WordPress installation on your local server.
2. **Install WooCommerce Plugin:**
    - Log in to your WordPress dashboard.
    - Navigate to Plugins > Add New.
    - Search for "WooCommerce," install, and activate the plugin.
3. **Configure WooCommerce:**
    - Follow the WooCommerce setup wizard to configure basic settings:
        1. Set up your store location, currency, and payment methods.
        2. Add shipping options.
        3. Payment Integration E-Sewa
    - Create at least 5 sample products with categories and subcategories, images, and descriptions.
4. **Enforce User Authentication Before Checkout:**
    - Install and activate the "WooCommerce Force Authenticate Before Checkout" plugin.
    - Configure the plugin to require users to log in or sign up before proceeding to check out.
    - Test the functionality by attempting to checkout without logging in.
5. **Customize the Website:**
    - Choose and install a WooCommerce-compatible theme (e.g. ThShop mania).
    - Customize the theme to match your store's branding (logo, colors, fonts, etc.).
    - Add necessary pages like, Landing Page, Shop page, Contact Us, and Privacy Policy.
    - Add 30 products
    - Add a Mega Menu for view categories and Sub-categories
    - Ensure a mobile-friendly design.
6. **Product & Category Setup**

- Add a total of 30 products to the store.
- Organize products under 6 main categories, each containing relevant shop sub-categories.
- Configure different product types, including:
    - Simple Products
    - Variable Products (with at least 3–4 variations, such as size, color, or weight).

7. **Test the Website:**

- Test the website functionality, including product browsing, adding to cart, and checkout process.
- Ensure the login/signup requirement works as expected.

Part 2: Transfer the Website from Localhost to a Live Server

1. **Export the Local Website:**
   - Use a plugin like All-in-One WP Migration or Duplicator to export your local WordPress site.
2. **Domain & Hosting Setup:**
   - Apply For Free Domain (E.g. .np Domain registrations)
   - Log in to your hosting provider's cPanel.
   - Create a new database and user for your WordPress site.
   - Upload the WordPress files to your public HTML directory or a subfolder.
3. **Import the Website:**
   - Use the same migration plugin to import the website files and database to the live server.
   - Update the wp-config.php file with the new database credentials.
4. **Test the Live Website:**
   - Visit your live website URL and ensure everything is working correctly.
   - Test the login/signup and checkout process again.

Part 3: Documentation and Submission

1. **Document of the Process:**
   - Write a step-by-step report explaining how you set up the website, including screenshots of key steps.
2. **Submission Guidelines:**

- Submit a report (PDF/Word) covering:
  - Theme & plugins used
  - Challenges faced and solutions
  - Screenshots of the admin panel and frontend.

3. **Submit the Assignment:**
   - Provide the live website URL.
   - Submit the documentation report PDF.

Share the WordPress admin credentials (for evaluation purposes only).

Evaluation and Marking Scheme:

- Content Relevance – **30%**
- Clarity & Organization – **20%**
- Depth of Analysis – **30%**
- Use of Visual Aids – **10%**
- Timely Submission – **10%**

Assignment Cover Page

Texas College of Management & IT

Bachelor of Information Technology (BIT)

Assignment Title: …….

Submitted By:

Name: _____

LCID: _____

Year/Semester: ……..

Submitted To:

Instructor Name: ……..

Course: …….

Date of Submission: _____

Texas College of Management & IT, Kathmandu

Assignment Structure

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